Hyperspectral and Multispectral Image Fusion via Superpixel-Based Weighted Nuclear Norm Minimization

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***Abstract*— Integrating a low-resolution hyperspectral image (HSI) and a high-resolution multispectral image is widely acknowledged as an effective approach for generating a high-resolution HSI. Recent studies have highlighted the nuclear norm as an efficient method for this problem through the utilization of low-rankness. However, the standard nuclear norm has a limitation due to treating singular values equally. To address this issue, we have incorporated the concept of the weighted nuclear norm (WNN) from the image denoising problem into HSI fusion, ensuring the retention of crucial data components. Furthermore, we propose a unified framework which integrates the WNN, a sparse prior, and total variation (TV) regularization. This framework utilizes the ℓ_1 norm of coefficients to promote spatial-spectral sparsity in the fused images, while TV is employed to preserve the spatial piecewise smooth structure. To efficiently solve the proposed model, we have designed an alternating direction method of multipliers (ADMM). The experimental results show that our proposed approach surpasses the state-of-the-art methods.**

***Index Terms*— Hyperspectral and multispectral image fusion, sparsity, superpixel, total variation (TV), weighted nuclear norm (WNN).**

I. Introduction

HYPERSPECTRAL imaging technology is a novel imaging technique that captures solar electromagnetic wave signals reflected by surface materials through airborne or spaceborne imaging spectrometers. The acquisition of multidimensional information by combining imaging and spectral technologies is referred to as hyperspectral remote sensing. Currently, hyperspectral remote sensing technology has emerged as a cutting-edge field in remote sensing and is extensively used in various domains, including disaster assessment [1], military reconnaissance [2], and other fields [3].

Despite the rapid advancements, hyperspectral imaging sensors are restricted by physical conditions and can only acquire hyperspectral images (HSIs) with low spatial resolution. This significantly restricts the applications and progress of HSIs in diverse fields. Conversely, superior spatial resolution multispectral images can be obtained by multispectral imaging sensors. Currently, an efficient method for reconstructing a high-resolution HSI (HR-HSI) is to merge a high-resolution multispectral image (HR-MSI) and a low-resolution HSI (LR-HSI) of the identical scene.

Recently, several HSI-MSI fusion methods can be divided into three categories: matrix factorization-based approaches, tensor factorization-based approaches, and deep learning-based approaches [4], [5], [6], [7]. In this study, we focus on fusion methods via matrix factorization. Kawakami et al. [8] first introduced matrix factorization to the fusion of LR-HSI and HR-RGB. They learned the spectral dictionary from the LR-HSI using a sparse prior and estimated the corresponding coefficients from the HR-RGB image. Wycoff et al. [9], presented a nonnegative sparse matrix factorization method that fully considered the sparsity of HSIs. To enhance computational efficiency, an alternating direction method of multipliers (ADMM) was employed to calculate the spectral dictionary and its coefficient matrix. Moreover, in [10], [11], [12], and [13], various coupled matrix factorization techniques were proposed that utilized spectral unmixing prior information for the purpose of regulating the HSI fusion problem. These methods have the advantage of updating the spectral dictionary and coefficients alternately. Specifically, Wei et al. [11] constructed the fusion problem as maximizing the joint posterior distribution of endmember signatures and abundance maps via a spectral unmixing method. This approach improved the estimation of endmember and abundance. Feng et al. [13] presented a new HSI fusion method by using the linear spectral mixture model. This method investigated the spatial relationship between the LR-HSI and the HR-MSI's endmember signatures, and could effectively fuse the spatial-spectral information.

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To effectively handle the HSI-MSI fusion problem, it is essential to incorporate spectral information with the spatial structure information of the images. To achieve spatial smoothness and promote spatial similarity, various regularization methods have been employed. For instance, Simões et al. [14] utilized total variation (TV) regularization, while Wei et al. [15] employed sparse regularization. In addition, Veganzones et al. [16] and Zhou et al. [12] learned the spectral dictionary for each patch and carried out sparse coding independently to account for local low-rank. Fang et al. [17] utilized similarity between pixels in each superpixel to improve sparse decomposition. Dian et al. [18] proposed a method that integrated sparse prior with local low-rank prior to construct a HSI super-resolution method. They used the superpixel segmentation method based on entropy rate to segment the HR-MSI. Recently, Xue et al. [19] proposed a novel HSI super-resolution method based on spatial-spectral structured sparse low-rank representation, which fully considered the spatial-spectral low-rank relationships between the available images (namely the LR-HSI and the HR-MSI) and the latent HR-HSI. Finally, a recent HSI fusion method combining TV regularization with sparse prior and local low-rank prior was presented to better preserve the spatial structure such as edges [20].

Recently, numerous methods using the low-rank constraint have emerged for image denoising and fusion [16], [18], [21], [22], [23] due to the low-rank nature of HSIs. Among them, nuclear norm minimization (NNM) [24], [25] is a popular approach for applying the low-rank constraint owing to its tight convex relaxation of the low-rank matrix factorization problem. However, NNM has some limitations, as it treats each singular value equally during soft-thresholding, shrinking each singular value by the same amount. This neglects the fact that larger singular values contain more vital data than smaller ones. To overcome this, different weights must be assigned to singular values, with larger singular values being shrunk less. To achieve this, Gu et al. [26] proposed the weighted nuclear norm (WNN), which provides improved flexibility. Gu et al. [27] further demonstrated that WNN minimization (WNNM) achieved state-of-the-art performance in common low-level vision tasks such as image denoising, image inpainting, and background subtraction.

In recent years, a variety of approaches based on the WNN have been proposed for HSI denoising and fusion [28], [29], [30], [31], [32], including HSI mixed denoising. The reason behind the popularity of WNN is its ability to leverage the low-rank property of HSIs. To further enhance the performance of WNN, it is often combined with the TV regularization which can preserve edge information and promote spatial smoothness. Recently, patch-based WNNM methods have been studied, which consider spatial nonlocal similarity to upgrade the performance of WNNM [33], [34], [35]. For instance, in [34], a tensor-based HSI denoising method was proposed which used the tensor WNNM to separate the low-rank clean HSI patches. Additionally, Kim et al. [35] presented a patch-based WNNM approach for removing additive Cauchy noise in images. In [36], a superpixel-based WNN method which integrated with the TV regularization was proposed for hyperspectral sparse unmixing. To improve the flexibility of tensor nuclear norm, Zhang et al. [37] proposed a nonlocal HSI denoising method via WNNs of transformed tensors. WNNM has also been used for dealing with color images. For example, Xu et al. [38] presented a multichannel WNNM approach for real color image denoising where a weight matrix was introduced to balance the noise in the RGB channels. Huang et al. [39] proposed the quaternion-based WNNM for color image restoration to reduce color artifacts. Currently, WNNM has attracted significant attention due to its competitive performance.

Building on the success of the WNN method in various applications, we extend its utility to HSI fusion in this study. Specifically, we propose a novel model for hyperspectral sparse fusion by integrating WNN and TV regularization. To obtain the fused image, we follow the standard practice of first learning the spectral dictionary from the LR-HSI, and then solving the coefficient matrix from both the LR-HSI and the HR-MSI. By adding the ℓ_1 norm of coefficients, we encourage the sparsity of the fused image under a suitable spectral dictionary. To better promote local low-rank, we integrate a weighted vector with the nuclear norm. Additionally, to preserve image edges and enhance structural smoothness, we combine the TV regularization with these two regularizations. To efficiently solve the proposed model, we use the ADMM and experiment with the CAVE dataset. Our method outperforms the current state-of-the-art matrix factorization approach by increasing the peak signal-to-noise ratio (PSNR) value by over 0.34 and reducing the ERGAS value by 6.7%.

The rest of the article is structured in the following way. We introduce the notation and discuss the fusion framework, as well as review two existing HSI-MSI fusion models that use superpixel-based NNM in Section II. Then, we present our proposed fusion model and describe its optimization algorithm in Section III. We also provide some insights on how to determine the weight vector. Experimental results and discussions are presented in Section IV. Finally, we conclude the article in Section V.

II. Related Works

A. Problem Formulation

To begin with, we introduce some notations and then present the fusion framework that is relevant to the HSI-MSI fusion problem. Hyperspectral and multispectral images are essentially 3-D data cubes that comprise 2-D spatial information and 1-D spectral information. However, we use 2-D matrices to represent HR-HSI, LR-HSI, and HR-MSI. The number of spectral bands is denoted by the first dimension of the matrix, while the number of pixels is represented by the second dimension. Specifically, HR-HSI is denoted by $\mathbf{X} \in \mathbb{R}^{S\times N}$, where S and N are respectively the numbers of spectral bands and pixels. LR-HSI is represented by $\mathbf{Y} \in \mathbb{R}^{S\times n}$, where n also denotes the number of pixels. Compared to $\mathbf{X}$, $\mathbf{Y}$ has a lower spatial resolution, resulting in $n < N$. Finally, $\mathbf{Z} \in \mathbb{R}^{s\times N}$ represents the HR-MSI in the same scene, where $s < S$ indicates that $\mathbf{Z}$ has fewer spectral bands than $\mathbf{X}$. Spatial downsampling of the HR-HSI $\mathbf{X}$ results in the acquisition of

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the LR-HSI $\mathbf{Y}$. As a result, $\mathbf{Y}$ can be represented as follows:

$$\mathbf{Y} = \mathbf{XBM} + \mathbf{N}_h \tag{1}$$

where $\mathbf{B} \in \mathbb{R}^{N\times N}$ stands for a blurring matrix that emulates the convolution operation of the hyperspectral imaging sensor. $\mathbf{M} \in \mathbb{R}^{N\times n}$ is a matrix used for spatial downsampling, while $\mathbf{N}_h$ denotes the Gaussian additive noise. In a similar manner, the HR-MSI $\mathbf{Z}$ within the same scene can be obtained by spectral downsampling of the HR-HSI $\mathbf{X}$. As a result, $\mathbf{Z}$ can be expressed as follows:

$$\mathbf{Z} = \mathbf{HX} + \mathbf{N}_m \tag{2}$$

where $\mathbf{H} \in \mathbb{R}^{s\times S}$ is a matrix used for spectral downsampling. Its rows represent the spectral response of the multispectral imaging sensor. Additionally, $\mathbf{N}_m$ denotes the Gaussian additive noise.

There exists a strong correlation among the spectral bands in HSIs. Consequently, each pixel in the spectral vector $\mathbf{x}_i$ of the HR-HSI can be represented as a linear combination of various spectral features [40]. In other words, we get

$$\mathbf{x}_i = \mathbf{E}\mathbf{a}_i. \tag{3}$$

Here, $\mathbf{E} \in \mathbb{R}^{S\times L}$ is a spectral dictionary, and $\mathbf{a}_i \in \mathbb{R}^L$ stands for the corresponding coefficients. In this way, we deduce that

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N] = \mathbf{EA} \tag{4}$$

where $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_N] \in \mathbb{R}^{L\times N}$ is a coefficient matrix. By integrating (4) into (1) and (2), $\mathbf{Y}$ and $\mathbf{Z}$ can be expressed as follows:

$$\mathbf{Y} = \mathbf{EABM} + \mathbf{N}_h \tag{5}$$

$$\mathbf{Z} = \mathbf{HEA} + \mathbf{N}_m. \tag{6}$$

By using this approach, the fusion problem is converted into the estimation of the spectral dictionary $\mathbf{E}$ and its corresponding coefficient matrix $\mathbf{A}$ from $\mathbf{Y}$ and $\mathbf{Z}$. Ultimately, the fused image can be obtained using (4).

B. Nuclear Norm Minimization

As stated earlier, the objective of the fusion process is to estimate the spectral dictionary $\mathbf{E}$ and the coefficient matrix $\mathbf{A}$ using (5) and (6). The initial step involves learning $\mathbf{E}$ from $\mathbf{Y}$ since it retains most of the spectral information of $\mathbf{X}$ [41]. To accomplish this, we must study a sparse nonnegative matrix decomposition problem [42]. The block coordinate descent optimization algorithm [43] is used to update one column of $\mathbf{E}$ while keeping the others constant. Once the spectral dictionary $\mathbf{E}$ is obtained, the fusion problem can be simplified to estimating the corresponding coefficient matrix $\mathbf{A}$. Given the spectral dictionary, the coefficient matrix $\mathbf{A}$ can be estimated by

$$\min_{\mathbf{A}} \|\mathbf{Y} - \mathbf{EABM}\|_F^2 + \lambda\|\mathbf{Z} - \mathbf{HEA}\|_F^2 \tag{7}$$

where $\|\cdot\|_F$ represents the Frobenius norm, and $\lambda > 0$ is a parameter used to maintain a balance between the two fidelity terms. Due to the absence of certain prior information, the problem (7) is highly underdetermined and lacks uniqueness in its solution. Therefore, regularization techniques are essential to obtain a meaningful solution. It is widely acknowledged that under a suitable spectral dictionary, the coefficient matrix $\mathbf{A}$ can be sparse. Incorporating a sparse prior is necessary to leverage the sparsity of spatial distribution of ground objects and the high correlation between spectra. The effectiveness of a sparse prior in various HSI reconstruction problems has been demonstrated in several studies [8], [9], [41]. The sparsity of coefficient vectors $\mathbf{a}_i$ can be well-characterized by ℓ_0 norm, but its minimization is NP-hard. Therefore, ℓ_1 norm is usually used to approximate ℓ_0 norm. Specifically, to promote the sparsity of the coefficient matrix $\mathbf{A}$, we use the definition of $\|\mathbf{A}\|_1 = \sum_{i=1}^{N} \|\mathbf{a}_i\|_1$.

Furthermore, local low-rankness is a common property of HSIs since spectral vectors within appropriately selected neighborhoods represent the same materials. To address this, researchers have turned to NNM, a convex surrogate of the low-rank matrix factorization problem. For instance, Dian et al. [18] promoted local low-rankness for HSI super-resolution via the nuclear norm. The HR-MSI contains a significant portion of the spatial information in the HR-HSI, allowing it to be segmented into J superpixels through the entropy rate-based superpixel segmentation method [44]. Correspondingly, the HR-HSI can be divided into J superpixels $\{\mathbf{X}^{(j)} = \mathbf{EA}^{(j)}\}_{j=1}^{J}$ based on their spatial positions. Finally, each superpixel is subject to nuclear norm regularization to promote its low-rankness. Hence, the fusion problem can be constructed as follows [18]:

$$\min_{\mathbf{A}} \|\mathbf{Y} - \mathbf{EABM}\|_F^2 + \lambda\|\mathbf{Z} - \mathbf{HEA}\|_F^2 + \eta_1\|\mathbf{A}\|_1 + \eta_2 \sum_{j=1}^{J} \|\mathbf{A}^{(j)}\|_* \tag{8}$$

where η_1, η_2 are two regularization parameters, and $\|\mathbf{A}^{(j)}\|_*$ stands for the nuclear norm of the coefficients in the jth superpixel.

In our previous work [20], we introduced TV regularization to (8) to promote structural smoothness and better preserve edge information. TV is a classical method that has been widely applied in various image processing problems [21], [29], [36]. It has been proven to be effective in edge-preserving when used for image denoising, particularly for piecewise constant images. However, TV also has several drawbacks, and although some new models have been proposed to address these issues [45], [46], new problems have emerged. Nonetheless, overall, TV is a reliable choice for leveraging spatial information. The model presented in [20] can be expressed by the following minimization problem:

$$\min_{\mathbf{A}} \|\mathbf{Y} - \mathbf{EABM}\|_F^2 + \lambda\|\mathbf{Z} - \mathbf{HEA}\|_F^2 + \eta_1\|\mathbf{A}\|_1 + \eta_2 \sum_{j=1}^{J} \|\mathbf{A}^{(j)}\|_* + \eta_3\|\nabla\mathbf{A}\|_1 \tag{9}$$

where η_3 is a regularization parameter, and ∇ is the gradient operator.

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III. Proposed WNNM Method

Currently, NNM is extensively employed in low-rank matrix approximation. However, the solution to the NNM problem necessitates performing singular value decomposition (SVD) on the pertinent matrix at each iteration [47]. While the standard nuclear norm assumes equal importance for every singular value to ensure the convexity of the model, it disregards the prior knowledge of singular values of the matrix. Specifically, it fails to consider that larger singular values contain more data information than smaller ones. To address this issue, it is necessary to assign different weights to different singular values to emphasize their differences. Therefore, we construct the following model for estimating the coefficients $\mathbf{A}$ by introducing weighting vectors

$$\min_{\mathbf{A}} \|\mathbf{Y}-\mathbf{EABM}\|_F^2+\lambda\|\mathbf{Z}-\mathbf{HEA}\|_F^2+\eta_1\|\mathbf{A}\|_1 + \eta_2\sum_{j=1}^{J}\|\mathbf{A}^{(j)}\|_{\boldsymbol{\omega}^{(j)},*}+\eta_3\|\nabla\mathbf{A}\|_1 \tag{10}$$

where we denote $\|\mathbf{A}^{(j)}\|_{\boldsymbol{\omega}^{(j)},*}$ as the weighted nuclear norm of the coefficients in the jth superpixel, $\boldsymbol{\omega}^{(j)}=[\omega_1^{(j)},\ldots,\omega_L^{(j)}]^T$ and $\omega_i^{(j)}\geq 0$ $(i=1,2,\ldots,L)$ is a nonnegative weight.

A. Algorithm

In this section, we introduce an ADMMs for solving the proposed model (10). The ADMM is an efficient method to solve problems involving nondifferentiable and high-order functional minimization with various applications [48].

Specifically, (10) can be decomposed into several subproblems via the ADMM, and then each subproblem can be solved analytically. First, six auxiliary variables $\mathbf{P}_1,\mathbf{P}_2,\mathbf{P}_3,\mathbf{P}_4,\mathbf{P}_5$ and $\mathbf{P}_6$ are introduced into the proposed model (10). Hence, it can be rewritten as the following constrained optimization problem

$$\begin{aligned}\min_{\mathbf{A},\mathbf{P}_1,\mathbf{P}_2,\mathbf{P}_3,\mathbf{P}_4,\mathbf{P}_5,\mathbf{P}_6}\ & \|\mathbf{Y}-\mathbf{EP}_1\mathbf{M}\|_F^2+\lambda\|\mathbf{Z}-\mathbf{HEP}_2\|_F^2 \\ & +\eta_1\|\mathbf{P}_3\|_1+\eta_2\sum_{j=1}^{J}\left\|\mathbf{P}_4^{(j)}\right\|_{\boldsymbol{\omega}^{(j)},*}+\eta_3\|\mathbf{P}_6\|_1 \\ \text{s.t. } & \mathbf{P}_1=\mathbf{AB},\quad \mathbf{P}_2=\mathbf{A},\ \mathbf{P}_3=\mathbf{A},\ \mathbf{P}_4=\mathbf{A} \\ & \mathbf{P}_5=\mathbf{A},\quad \mathbf{P}_6=\nabla\mathbf{P}_5.\end{aligned} \tag{11}$$

To solve problem (11), six Lagrangian multipliers $\mathbf{G}_1,\mathbf{G}_2,\mathbf{G}_3,\mathbf{G}_4,\mathbf{G}_5$ and $\mathbf{G}_6$ are utilized. The corresponding augmented Lagrangian function is

$$\begin{aligned}&\mathcal{L}(\mathbf{A},\mathbf{P}_1,\mathbf{P}_2,\mathbf{P}_3,\mathbf{P}_4,\mathbf{P}_5,\mathbf{P}_6;\mathbf{G}_1,\mathbf{G}_2,\mathbf{G}_3,\mathbf{G}_4,\mathbf{G}_5,\mathbf{G}_6) \\ &=\|\mathbf{Y}-\mathbf{EP}_1\mathbf{M}\|_F^2+\mu\|\mathbf{P}_1-\mathbf{AB}+\mathbf{G}_1\|_F^2 \\ &\quad+\lambda\|\mathbf{Z}-\mathbf{HEP}_2\|_F^2+\mu\|\mathbf{P}_2-\mathbf{A}+\mathbf{G}_2\|_F^2+\eta_1\|\mathbf{P}_3\|_1 \\ &\quad+\mu\|\mathbf{P}_3-\mathbf{A}+\mathbf{G}_3\|_F^2+\eta_2\sum_{j=1}^{J}\left\|\mathbf{P}_4^{(j)}\right\|_{\boldsymbol{\omega}^{(j)},*} \\ &\quad+\mu\|\mathbf{P}_4-\mathbf{A}+\mathbf{G}_4\|_F^2+\mu\|\mathbf{P}_5-\mathbf{A}+\mathbf{G}_5\|_F^2 \\ &\quad+\eta_3\|\mathbf{P}_6\|_1+\mu\|\mathbf{P}_6-\nabla\mathbf{P}_5+\mathbf{G}_6\|_F^2\end{aligned} \tag{12}$$

where $\mu>0$ is a constant. When using the ADMM, we need to perform alternating iterations on seven subproblems and update six Lagrangian multipliers at the same time. That is, we have (13), as shown at the bottom of the next page.

Next, the $\mathbf{A}$-subproblem can be written as

$$\begin{aligned}\mathbf{A}^{k+1}=\arg\min_{\mathbf{A}}\ & \left\|\mathbf{P}_1^k-\mathbf{AB}+\mathbf{G}_1^k\right\|_F^2 \\ &+\left\|\mathbf{P}_2^k-\mathbf{A}+\mathbf{G}_2^k\right\|_F^2+\left\|\mathbf{P}_3^k-\mathbf{A}+\mathbf{G}_3^k\right\|_F^2 \\ &+\left\|\mathbf{P}_4^k-\mathbf{A}+\mathbf{G}_4^k\right\|_F^2+\left\|\mathbf{P}_5^k-\mathbf{A}+\mathbf{G}_5^k\right\|_F^2.\end{aligned} \tag{14}$$

By solving its Euler–Lagrange equation, we have

$$\begin{aligned}\mathbf{A}^{k+1}=&\left(\left(\mathbf{P}_1^k+\mathbf{G}_1^k\right)\mathbf{B}^T+\mathbf{P}_2^k+\mathbf{G}_2^k+\mathbf{P}_3^k+\mathbf{G}_3^k\right. \\ &\left.+\mathbf{P}_4^k+\mathbf{G}_4^k+\mathbf{P}_5^k+\mathbf{G}_5^k\right)\times\left[4\mathbf{I}+\mathbf{BB}^T\right]^{-1}\end{aligned} \tag{15}$$

where $\mathbf{I}$ is the identity matrix.

The $\mathbf{P}_1$-subproblem can be expressed as

$$\mathbf{P}_1^{k+1}=\arg\min_{\mathbf{P}_1}\|\mathbf{Y}-\mathbf{EP}_1\mathbf{M}\|_F^2+\mu\left\|\mathbf{P}_1-\mathbf{A}^{k+1}\mathbf{B}+\mathbf{G}_1^k\right\|_F^2. \tag{16}$$

Although $\mathbf{P}_1$ cannot be directly optimized, its solution can be obtained by solving $\mathbf{P}_1\mathbf{M}$ and $\mathbf{P}_1\overline{\mathbf{M}}$, where $\overline{\mathbf{M}}$ is a matrix that selects the pixels that are not chosen by the downsampling matrix $\mathbf{M}$. As a result, we get

$$\begin{cases}\mathbf{P}_1^{k+1}\mathbf{M}=\left(\mathbf{E}^T\mathbf{E}+\mu\mathbf{I}\right)^{-1}\left[\mathbf{E}^T\mathbf{Y}+\mu\left(\mathbf{A}^{k+1}\mathbf{B}-\mathbf{G}_1^k\right)\mathbf{M}\right] \\ \mathbf{P}_1^{k+1}\overline{\mathbf{M}}=\left(\mathbf{A}^{k+1}\mathbf{B}-\mathbf{G}_1^k\right)\overline{\mathbf{M}}.\end{cases} \tag{17}$$

The $\mathbf{P}_2$-subproblem can be formulated as

$$\mathbf{P}_2^{k+1}=\arg\min_{\mathbf{P}_2}\lambda\|\mathbf{Z}-\mathbf{HEP}_2\|_F^2+\mu\left\|\mathbf{P}_2-\mathbf{A}^{k+1}+\mathbf{G}_2^k\right\|_F^2. \tag{18}$$

Then, based on its optimality condition, we obtain

$$\mathbf{P}_2^{k+1}=[\lambda(\mathbf{HE})^T\mathbf{HE}+\mu\mathbf{I}]^{-1}\left(\lambda(\mathbf{HE})^T\mathbf{Z}+\mu\left(\mathbf{A}^{k+1}-\mathbf{G}_2^k\right)\right). \tag{19}$$

The $\mathbf{P}_3$-subproblem can be written as

$$\mathbf{P}_3^{k+1}=\arg\min_{\mathbf{P}_3}\eta_1\|\mathbf{P}_3\|_1+\mu\left\|\mathbf{P}_3-\mathbf{A}^{k+1}+\mathbf{G}_3^k\right\|_F^2. \tag{20}$$

This subproblem has a closed-form solution, which can be acquired by the soft-shrinkage operator. The specific form is as follows:

$$\mathbf{P}_3^{k+1}=\operatorname{soft}\left(\mathbf{A}^{k+1}-\mathbf{G}_3^k,\frac{\eta_1}{2\mu}\right) \tag{21}$$

where $\operatorname{soft}(x,\ y)=\operatorname{sign}(x)*\max(|x|-y,\ 0)$ denotes a soft-shrinkage function.

The $\mathbf{P}_4$-subproblem can be formulated as

$$\mathbf{P}_4^{k+1}=\arg\min_{\mathbf{P}_4}\eta_2\sum_{j=1}^{J}\left\|\mathbf{P}_4^{(j)}\right\|_{(\boldsymbol{\omega}^{k+1})^{(j)},*}+\mu\left\|\mathbf{P}_4-\mathbf{A}^{k+1}+\mathbf{G}_4^k\right\|_F^2. \tag{22}$$

For this subproblem, we need to solve each superpixel separately. For the jth superpixel $(\mathbf{P}_4^{k+1})^{(j)}$, we get the optimal solution

$$\left(\mathbf{P}_4^{k+1}\right)^{(j)}=\mathbf{U}S_{(\boldsymbol{\omega}^{k+1})^{(j)}}(\boldsymbol{\Sigma})\mathbf{V}^T \tag{23}$$

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where $\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ is the SVD of $(\mathbf{A}^{k+1})^{(j)} - (\mathbf{G}_4^k)^{(j)}$, and $S_{(\boldsymbol{\omega}^{k+1})^{(j)}}(\mathbf{\Sigma})$ denotes the generalized soft-thresholding operator with weight vector $(\boldsymbol{\omega}^{k+1})^{(j)}$. Specifically,

$$[S_{(\boldsymbol{\omega}^{k+1})^{(j)}}(\mathbf{\Sigma})]_{ii} = \max\left(\mathbf{\Sigma}_{ii} - \frac{\eta_2}{2\mu}\left(\omega_i^{k+1}\right)^{(j)}, 0\right) \tag{24}$$

where $\mathbf{\Sigma}_{ii}$ represents the ith element on the main diagonal of the diagonal matrix $\mathbf{\Sigma}$, and $(\omega_i^{k+1})^{(j)}$ denotes the ith element of the weighting vector for the jth superpixel in step $k+1$.

The $\mathbf{P}_5$-subproblem can be expressed as

$$\mathbf{P}_5^{k+1} = \arg\min_{\mathbf{P}_5} \left\|\mathbf{P}_5 - \mathbf{A}^{k+1} + \mathbf{G}_5^k\right\|_F^2 + \left\|\mathbf{P}_6^k - \nabla\mathbf{P}_5 + \mathbf{G}_6^k\right\|_F^2. \tag{25}$$

To effectively solve this subproblem, we use the fast Fourier transform (FFT) denoted as $\mathcal{F}$ under periodic boundary conditions. It follows that:

$$\mathbf{P}_5^{k+1} = \mathcal{F}^{-1}\left[\frac{\mathcal{F}\left[\left(\mathbf{A}^{k+1} - \mathbf{G}_5^k + \nabla^*\left(\mathbf{P}_6^k + \mathbf{G}_6^k\right)\right)\right]}{\mathcal{F}[-\Delta + \mathbf{I}]}\right] \tag{26}$$

where Δ is the Laplace operator, while $\mathcal{F}^{-1}$ is FFT's inverse transform.

The $\mathbf{P}_6$-subproblem is as follows:

$$\mathbf{P}_6^{k+1} = \arg\min_{\mathbf{P}_6} \eta_3\|\mathbf{P}_6\|_1 + \mu\left\|\mathbf{P}_6 - \nabla\mathbf{P}_5^{k+1} + \mathbf{G}_6^k\right\|_F^2. \tag{27}$$

Similarly, via the soft-shrinkage operator, we have

$$\mathbf{P}_6^{k+1} = \text{soft}\left(\nabla\mathbf{P}_5^{k+1} - \mathbf{G}_6^k, \frac{\eta_3}{2\mu}\right) \tag{28}$$

where $\text{soft}(\cdot,\cdot)$ signifies the soft-shrinkage operator as above.

The above steps are iterated until convergence for solving the proposed model (10).

B. Selection of the Weight Vector

Obviously, the key problem now is how to determine the weight vector $\boldsymbol{\omega}$. Originally, the WNN was proposed by Gu et al. [26] to apply to image denoising so as to highlight the importance of larger singular values. Specifically, by using the block matching method, the nonlocal similar patches across the image can be found and then stacked into a matrix called a superpixel. And the superpixel-based WNN model for image denoising is as follows:

$$\hat{\mathbf{u}}^{(j)} = \arg\min_{\mathbf{u}^{(j)}} \frac{1}{2}\|\mathbf{f}^{(j)} - \mathbf{u}^{(j)}\|_F^2 + \|\mathbf{u}^{(j)}\|_{\boldsymbol{\omega}^{(j)},*} \tag{29}$$

where $\mathbf{f}^{(j)}$ represents the jth superpixel of the noisy image $\mathbf{f}$.

The significance of the larger singular values of $\mathbf{u}^{(j)}$ being greater than the smaller ones implies that less shrinkage should be applied as the singular value increases. Therefore, to achieve this, the weight assigned to the ith singular value of $\mathbf{u}^{(j)}$, denoted as $\sigma_i(\mathbf{u}^{(j)})$, should be inversely proportional to its value. In their work, the weight vector $\boldsymbol{\omega}$ was denoted by

$$\left(\omega_i^{k+1}\right)^{(j)} = \frac{c\sqrt{n}}{\sigma_i[(\mathbf{u}^k)^{(j)}] + \varepsilon} \tag{30}$$

where n is the number of similar patches in $\mathbf{f}^{(j)}$, $c > 0$ is a positive constant, and ε is a small constant to avoid singularity.

In the following, we explore setting the weight vector for HSI-MSI fusion. Based on the above observation, considering the $\mathbf{P}_4$ subproblem

$$\mathbf{P}_4^{k+1} = \arg\min_{\mathbf{P}_4} \eta_2 \sum_{j=1}^{J} \left\|\mathbf{P}_4^{(j)}\right\|_{(\boldsymbol{\omega}^{k+1})^{(j)},*} + \mu\|\mathbf{P}_4 - \mathbf{A}^{k+1} + \mathbf{G}_4^k\|_F^2. \tag{31}$$

The weight vector $\boldsymbol{\omega}$ can be similarly estimated as

$$\left(\omega_i^{k+1}\right)^{(j)} = \frac{\sqrt{b^{(j)}}}{\sigma_i\left[\left(\mathbf{P}_4^k\right)^{(j)}\right] + \varepsilon} \tag{32}$$

where $b^{(j)}$ is the number of similar patches for the jth superpixel, $\sigma_i[(\mathbf{P}_4^k)^{(j)}]$ is the ith singular value of $(\mathbf{P}_4^k)^{(j)}$,

$$\begin{cases}
\mathbf{A}^{k+1} = \arg\min_{\mathbf{A}} \mathcal{L}\left(\mathbf{A}, \mathbf{P}_1^k, \mathbf{P}_2^k, \mathbf{P}_3^k, \mathbf{P}_4^k, \mathbf{P}_5^k, \mathbf{P}_6^k; \mathbf{G}_1^k, \mathbf{G}_2^k, \mathbf{G}_3^k, \mathbf{G}_4^k, \mathbf{G}_5^k, \mathbf{G}_6^k\right) \\
\mathbf{P}_1^{k+1} = \arg\min_{\mathbf{P}_1} \mathcal{L}\left(\mathbf{A}^{k+1}, \mathbf{P}_1, \mathbf{P}_2^k, \mathbf{P}_3^k, \mathbf{P}_4^k, \mathbf{P}_5^k, \mathbf{P}_6^k; \mathbf{G}_1^k, \mathbf{G}_2^k, \mathbf{G}_3^k, \mathbf{G}_4^k, \mathbf{G}_5^k, \mathbf{G}_6^k\right) \\
\mathbf{P}_2^{k+1} = \arg\min_{\mathbf{P}_2} \mathcal{L}\left(\mathbf{A}^{k+1}, \mathbf{P}_1^{k+1}, \mathbf{P}_2, \mathbf{P}_3^k, \mathbf{P}_4^k, \mathbf{P}_5^k, \mathbf{P}_6^k; \mathbf{G}_1^k, \mathbf{G}_2^k, \mathbf{G}_3^k, \mathbf{G}_4^k, \mathbf{G}_5^k, \mathbf{G}_6^k\right) \\
\mathbf{P}_3^{k+1} = \arg\min_{\mathbf{P}_3} \mathcal{L}\left(\mathbf{A}^{k+1}, \mathbf{P}_1^{k+1}, \mathbf{P}_2^{k+1}, \mathbf{P}_3, \mathbf{P}_4^k, \mathbf{P}_5^k, \mathbf{P}_6^k; \mathbf{G}_1^k, \mathbf{G}_2^k, \mathbf{G}_3^k, \mathbf{G}_4^k, \mathbf{G}_5^k, \mathbf{G}_6^k\right) \\
\mathbf{P}_4^{k+1} = \arg\min_{\mathbf{P}_4} \mathcal{L}\left(\mathbf{A}^{k+1}, \mathbf{P}_1^{k+1}, \mathbf{P}_2^{k+1}, \mathbf{P}_3^{k+1}, \mathbf{P}_4, \mathbf{P}_5^k, \mathbf{P}_6^k; \mathbf{G}_1^k, \mathbf{G}_2^k, \mathbf{G}_3^k, \mathbf{G}_4^k, \mathbf{G}_5^k, \mathbf{G}_6^k\right) \\
\mathbf{P}_5^{k+1} = \arg\min_{\mathbf{P}_5} \mathcal{L}\left(\mathbf{A}^{k+1}, \mathbf{P}_1^{k+1}, \mathbf{P}_2^{k+1}, \mathbf{P}_3^{k+1}, \mathbf{P}_4^{k+1}, \mathbf{P}_5, \mathbf{P}_6^k; \mathbf{G}_1^k, \mathbf{G}_2^k, \mathbf{G}_3^k, \mathbf{G}_4^k, \mathbf{G}_5^k, \mathbf{G}_6^k\right) \\
\mathbf{P}_6^{k+1} = \arg\min_{\mathbf{P}_6} \mathcal{L}\left(\mathbf{A}^{k+1}, \mathbf{P}_1^{k+1}, \mathbf{P}_2^{k+1}, \mathbf{P}_3^{k+1}, \mathbf{P}_4^{k+1}, \mathbf{P}_5^{k+1}, \mathbf{P}_6; \mathbf{G}_1^k, \mathbf{G}_2^k, \mathbf{G}_3^k, \mathbf{G}_4^k, \mathbf{G}_5^k, \mathbf{G}_6^k\right) \\
\mathbf{G}_1^{k+1} = \mathbf{P}_1^{k+1} - \mathbf{A}^{k+1}\mathbf{B} + \mathbf{G}_1^k \\
\mathbf{G}_2^{k+1} = \mathbf{P}_2^{k+1} - \mathbf{A}^{k+1} + \mathbf{G}_2^k \\
\mathbf{G}_3^{k+1} = \mathbf{P}_3^{k+1} - \mathbf{A}^{k+1} + \mathbf{G}_3^k \\
\mathbf{G}_4^{k+1} = \mathbf{P}_4^{k+1} - \mathbf{A}^{k+1} + \mathbf{G}_4^k \\
\mathbf{G}_5^{k+1} = \mathbf{P}_5^{k+1} - \mathbf{A}^{k+1} + \mathbf{G}_5^k \\
\mathbf{G}_6^{k+1} = \mathbf{P}_6^{k+1} - \nabla\mathbf{P}_5^{k+1} + \mathbf{G}_6^k
\end{cases} \tag{13}$$

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and $\varepsilon = 10^{-8}$. However, the experimental results show no obvious improvement which can be shown in Table I. Thus, this weighting method is not suitable for our model. So, we further change the weight vector to

$$\left(\omega_i^{k+1}\right)^{(j)} = \frac{\sqrt{b^{(j)}}}{\sigma_i\left[\left(\mathbf{A}^{k+1}\right)^{(j)} - \left(\mathbf{G}_4^k\right)^{(j)}\right] + \varepsilon}. \tag{33}$$

Its advantage is that the weight vector is updated via the current $\mathbf{A}^{k+1}$. We demonstrate the efficiency of this weighting scheme in Section IV-B.

The whole algorithm for WNNM-based HSI-MSI fusion is summarized in Algorithm 1.

Algorithm 1 WNNM-Based HSI-MSI Fusion

Input: $\mathbf{Y}, \mathbf{Z}$

learn the spectral dictionary $\mathbf{E}$ from $\mathbf{Y}$
Parameters: $\lambda, \eta_1, \eta_2, \eta_3, \mu$, kMax, and $\epsilon \in \mathbb{R}$
Initialization: $\mathbf{A}^0, \mathbf{P}_1^0, \mathbf{P}_2^0, \mathbf{P}_3^0, \mathbf{P}_4^0, \mathbf{P}_5^0, \mathbf{P}_6^0, \mathbf{G}_1^0, \mathbf{G}_2^0, \mathbf{G}_3^0, \mathbf{G}_4^0, \mathbf{G}_5^0, \mathbf{G}_6^0$, and $k = 0$
while $k <$kMax **do**
  Update $\mathbf{A}^{k+1}$ via (15)
  Update $\mathbf{P}_1^{k+1}$ via (17)
  Update $\mathbf{P}_2^{k+1}$ via (19)
  Update $\mathbf{P}_3^{k+1}$ via (21)
  Update $\mathbf{P}_4^{k+1}$ via (23)
  Update $\mathbf{P}_5^{k+1}$ via (26)
  Update $\mathbf{P}_6^{k+1}$ via (28)
  Update Lagrangian multipliers by the last six equations of (13)
  Assign k by $k+1$
  If $\|\mathbf{A}^k - \mathbf{A}^{k-1}\|_F / \|\mathbf{A}^k\|_F \le \epsilon$, stop the iteration
end while
return $\mathbf{A}^* = \mathbf{A}^n$
$\mathbf{X} = \mathbf{E}\mathbf{A}^*$

Output: $\mathbf{X}$

IV. Numerical Experiments

First, we introduce the dataset used in our experiments and present four widely used quantitative metrics for evaluating the quality of fusion results. The influence of parameters on the experimental results will be analyzed for providing the optimal parameters. Then, we will discuss the issue of how to select the weighting vectors. Finally, we will present a detailed report of the experimental results of our proposed method (WNNSTV). To demonstrate the superiority of our method, we will compare it with four other advanced methods, including HySure [14], LRSR [18], HSI super-resolution approach proposed in [49] (hereafter called NSSJS), and HSI fusion method proposed in [20] (referred to as SLRTV1). We use relative errors to measure the iteration process. The relative error in $\mathbf{A}^k$ is given by

$$R(\mathbf{A}^k) = \frac{\|\mathbf{A}^k - \mathbf{A}^{k-1}\|_F}{\|\mathbf{A}^k\|_F}. \tag{34}$$

When $R(\mathbf{A}^k) \le 10^{-3}$ or the iteration reaches 250, we stop the iteration. All numerical experiments were conducted using MATLAB (R2018a) on a desktop computer with a 2.90 GHz Intel[1] Core[2] i7-7500U CPU and 12 GB RAM.

A. Data and Experimental Settings

In our experiments, we utilize the widely used CAVE dataset [50], which is a ground-based HSI dataset commonly employed in HSI-MSI fusion [8], [18], [20], [42]. The dataset comprises 32 high-resolution HSIs, each with a size of $512 \times 512 \times 31$.

For our experiments, our test images include "Peppers," "Superballs," "Lemon slices," and "Flowers." In order to get the corresponding LR-HSI and HR-MSI as our observed data, we downsample the HR-HSI in both spatial and spectral dimensions. Specifically, a 7×7 symmetric Gaussian filter with a standard deviation of 2 is utilized to convolve HR-HSI. Then, we downsample every four pixels in both spatial dimensions for each HSI band. It results in a $128 \times 128 \times 31$ LR-HSI. Additionally, HR-MSI is obtained by downsampling HR-HSI with the spectral response of a Nikon D700 camera, resulting in a $512 \times 512 \times 3$ HR-MSI.

To account for noise, we introduce Gaussian noise with signal-to-noise ratios (SNRs) of 30 and 40 dB to the LR-HSI and HR-MSI, respectively.

Here, we introduce four quantitative metrics for evaluating the performance of different fusion models. The first index is the average PSNR [51]. It is expressed as the average PSNR of all bands for HSI

$$\mathrm{PSNR}(\widehat{\mathbf{X}}, \mathbf{X}) = \frac{1}{S}\sum_{i=1}^{S} \mathrm{PSNR}(\widehat{\mathbf{X}}^i, \mathbf{X}^i) \tag{35}$$

where the ith band image of the ground truth $\mathbf{X} \in \mathbb{R}^{S\times N}$ is represented by $\mathbf{X}^i$, while the ith band image of the reconstructed image $\widehat{\mathbf{X}} \in \mathbb{R}^{S\times N}$ is represented by $\widehat{\mathbf{X}}^i$. The PSNR is an objective quantitative index that is commonly used to evaluate image distortion, where a higher PSNR indicates smaller image distortion.

The universal image quality index (UIQI) [52] is defined as

$$\mathrm{UIQI}(\widehat{\mathbf{X}}, \mathbf{X}) = \frac{1}{S}\sum_{i=1}^{S} \mathrm{Q}(\widehat{\mathbf{X}}^i, \mathbf{X}^i) \tag{36}$$

where $\mathrm{Q}(\widehat{\mathbf{X}}^i, \mathbf{X}^i)$ represents the UIQI between ith band images $\widehat{\mathbf{X}}^i$ and $\mathbf{X}^i$. The specific expression is as follows:

$$\mathrm{Q}(\widehat{\mathbf{X}}^i, \mathbf{X}^i) = \frac{1}{M}\sum_{j=1}^{M} \frac{\sigma_{\mathbf{X}_j^i\widehat{\mathbf{X}}_j^i}}{\sigma_{\mathbf{X}_j^i}\sigma_{\widehat{\mathbf{X}}_j^i}} \frac{2\mu_{\mathbf{X}_j^i}\mu_{\widehat{\mathbf{X}}_j^i}}{\mu_{\mathbf{X}_j^i}^2 + \mu_{\widehat{\mathbf{X}}_j^i}^2} \frac{2\sigma_{\mathbf{X}_j^i}\sigma_{\widehat{\mathbf{X}}_j^i}}{\sigma_{\mathbf{X}_j^i}^2 + \sigma_{\widehat{\mathbf{X}}_j^i}^2}$$

where the size of each window is 32×32 and there are M windows. The ground truth image and the reconstructed image of the ith band of the jth window are denoted by $\mathbf{X}_j^i$ and $\widehat{\mathbf{X}}_j^i$, respectively. The standard deviations of $\mathbf{X}_j^i$ and $\widehat{\mathbf{X}}_j^i$ are represented by $\sigma_{\mathbf{X}_j^i}$ and $\sigma_{\widehat{\mathbf{X}}_j^i}$, respectively, and their mean values are $\mu_{\mathbf{X}_j^i}$ and $\mu_{\widehat{\mathbf{X}}_j^i}$. Furthermore, $\sigma_{\mathbf{X}_j^i\widehat{\mathbf{X}}_j^i}$ denotes the covariance between them. The UIQI is a quality index measuring the similarity between the reconstructed image and the ground

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TABLE I
COMPARISON OF THE DIFFERENT WEIGHTING SCHEMES WITH TWO TESTING IMAGES

Image	Method	PSNR	UIQI	ERGAS	SAM
Superballs	WNN1	47.023	0.853	2.696	7.720
	WNN2	**48.358**	**0.872**	**2.342**	**7.098**
Flowers	WNN1	46.360	0.757	2.007	10.942
	WNN2	**48.132**	**0.778**	**1.628**	**9.675**

TABLE II
PSNR VALUES ABOUT DIFFERENT NUMBERS OF SUPERPIXELS J

Image	Method	1	100	200	300	400
Peppers	LRSR	46.07	47.39	47.46	47.41	47.47
	SLRTV1	46.99	47.74	47.78	47.74	47.73
	WNNSTV	-	48.25	48.33	48.33	48.36
Superballs	LRSR	46.74	47.76	47.77	47.73	47.74
	SLRTV1	47.26	47.91	47.90	47.90	47.91
	WNNSTV	-	48.30	48.36	48.34	48.32

TABLE III
QUANTITATIVE EVALUATION OF THE DIFFERENT FUSION ALGORITHMS WITH FOUR IMAGES

Image	Method	PSNR	UIQI	ERGAS	SAM
Peppers	HySure	45.887	0.855	2.365	7.036
	LRSR	47.459	0.855	2.181	6.609
	NSSJS	47.561	0.855	2.129	6.195
	SLRTV1	47.775	0.859	2.091	6.955
	WNNSTV	**48.329**	**0.868**	**1.888**	**5.886**
Superballs	HySure	46.409	0.853	2.766	8.209
	LRSR	47.766	0.867	2.566	7.346
	NSSJS	47.292	0.863	2.673	7.411
	SLRTV1	47.897	0.869	2.509	7.361
	WNNSTV	**48.358**	**0.872**	**2.342**	**7.098**
Lemon slices	HySure	44.278	0.866	2.809	9.971
	LRSR	46.581	0.884	2.440	7.073
	NSSJS	45.711	0.883	2.563	7.091
	SLRTV1	46.614	0.884	2.446	7.091
	WNNSTV	**47.000**	**0.888**	**2.282**	**6.718**
Flowers	HySure	45.758	0.768	2.076	12.608
	LRSR	47.652	0.768	1.741	10.887
	NSSJS	47.428	0.761	1.870	10.786
	SLRTV1	47.798	0.774	1.751	10.999
	WNNSTV	**48.132**	**0.778**	**1.628**	**9.675**

truth image, ranging from 0 to 1. A value closer to 1 indicates that the fused image is closer to the ground truth image.

Another metric is the relative dimensionless global error in synthesis (ERGAS) [53]. It calculates the spectral distortion in the image, which can be expressed as

$$\mathrm{ERGAS}(\widehat{\mathbf{X}}, \mathbf{X}) = \frac{100}{d}\sqrt{\frac{1}{S}\sum_{i=1}^{S}\frac{\mathrm{mse}(\widehat{\mathbf{X}}^i, \mathbf{X}^i)}{\mu_{\widehat{\mathbf{X}}^i}^2}} \tag{37}$$

where d represents a spatial downsampling factor, and the mean value of $\widehat{\mathbf{X}}^i$ is denoted as $\mu_{\widehat{\mathbf{X}}^i}$. Here, $\mathrm{mse}(\widehat{\mathbf{X}}^i, \mathbf{X}^i)$ stands for the mean squared error of $\widehat{\mathbf{X}}^i$ and $\mathbf{X}^i$. The smaller its value is, the better the spectral quality of the reconstructed image is within the spectral range.

The last metric used for quantitatively analyzing spectral information in each pixel is the spectral angle mapper (SAM) [54]. SAM measures the average angle between the ground truth image pixel $\mathbf{x}_j$ and the corresponding pixel in the reconstructed image $\hat{\mathbf{x}}_j$. Its definition can be expressed as follows:

$$\mathrm{SAM}(\widehat{\mathbf{X}}, \mathbf{X}) = \frac{1}{N}\sum_{j=1}^{N}\arccos\left(\frac{\widehat{\mathbf{x}}_j^T\mathbf{x}_j}{\|\widehat{\mathbf{x}}_j\|_2\|\mathbf{x}_j\|_2}\right). \tag{38}$$

SAM measures the similarity between spectra by quantifying the angle between two vectors. The smaller the angle between the vectors, the more similar the two spectra are. In other words, a smaller value of SAM indicates a better image fusion result.

Our proposed method involves seven parameters, including the number of atoms in the dictionary, which we set to 24, as in SLRTV1. We also explore the effect of the number of superpixels, denoted by J, on the fusion results. When J is set to a small value, the segmentation of the HR-MSI produces large superpixels, which may not fully utilize the similarity of neighboring pixels, leading to biased fusion results. Conversely, as J increases, the size of each superpixel decreases, making the pixels within a superpixel more similar. However, larger J values increase the encoding times of the algorithm. We evaluate the PSNR of two images under different numbers of superpixels, as shown in Table II. Our observations indicate that a J value less than 200 leads to relatively low PSNR values. As the value of J increases beyond 200, the PSNR value becomes more stable. Therefore, we set J to 200 in our experiments.

The other five parameters $\lambda, \eta_1, \eta_2, \eta_3, \mu$ need to be adjusted. In general, the fusion results mainly depend on the model parameters $\lambda, \eta_1, \eta_2, \eta_3$. Among them, λ is used to control the balance between the two data-fidelity terms. The regularization parameters η_1, η_2, η_3 play a significant role in determining the trade-off between the data-fidelity terms and the regularization terms. To be more precise, when the values of the regularization parameters are too big, the resulting fused image will be excessively smoothed, leading to a loss of some details. Conversely, if their values are too small, a nonsmooth fused result with blur will be generated. Therefore, their values need to be chosen according to the image structures. μ is the penalty parameter, which only controls the convergence rate but not the fusion result. In practice, we began by sampling several combinations of these parameters within their respective value ranges. Through a process of trial and error, we gradually adjusted the parameters to determine the optimal parameter set that produced the highest PSNR value.

In the test image "Peppers," we set the corresponding parameters as $\lambda = 0.6, \eta_1 = 1.5 \cdot 10^{-4}, \eta_2 = 2.1 \cdot 10^{-4}, \eta_3 = 6.5 \cdot 10^{-5}, \mu = 2.5 \cdot 10^{-3}$. Regarding the image "Superballs," the parameters are selected as: $\lambda = 0.5, \eta_1 = 1.2 \cdot 10^{-4}$,

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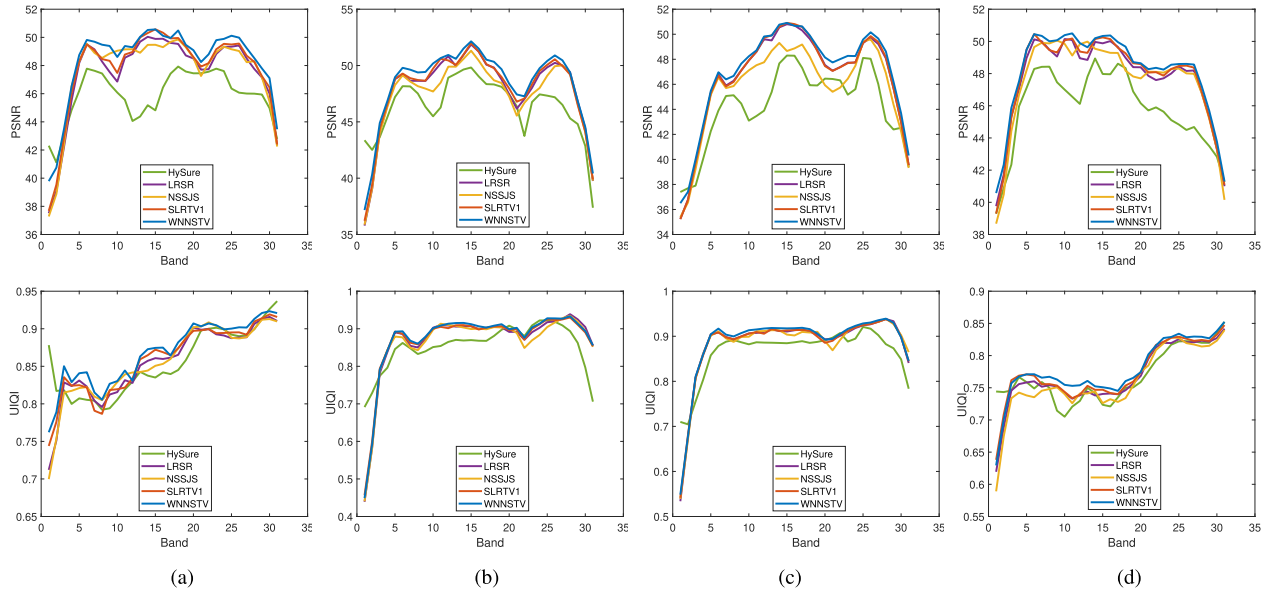


Fig. 1. (Top) Plots of PSNR and (bottom) UIQI values versus the spectral band. (a) Peppers. (b) Superballs. (c) Lemon slices. (d) Flowers.

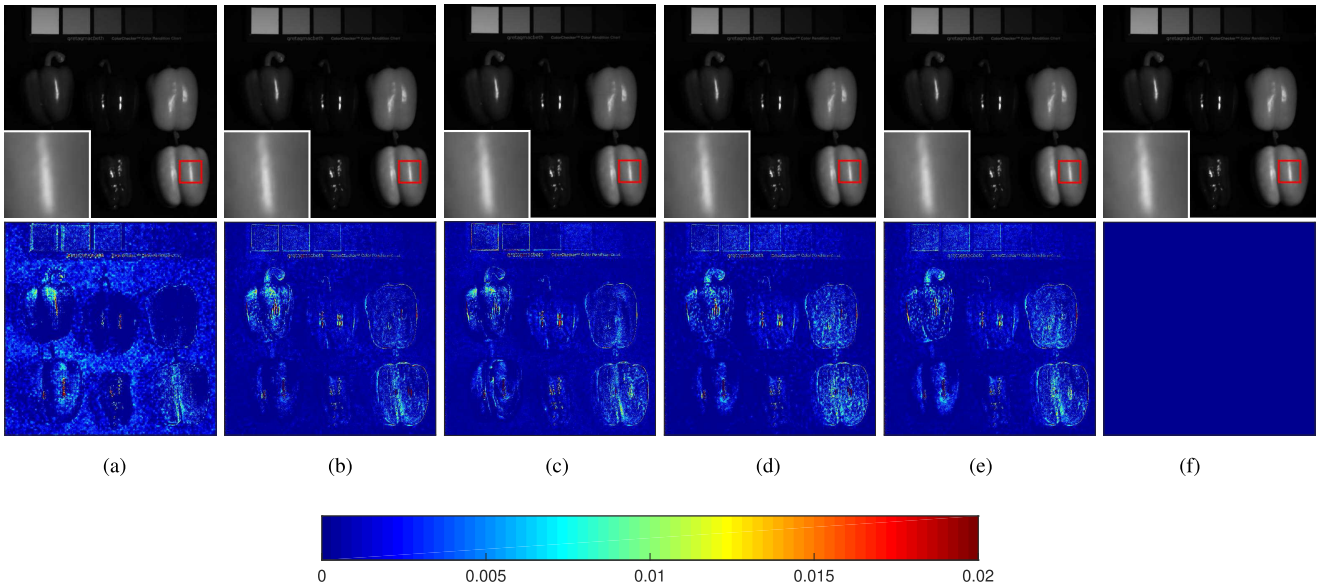


Fig. 2. Fused and error images of "Peppers" at the 20th spectral band. (a) HySure. (b) LRSR. (c) NSSJS. (d) SLRTV1. (e) WNNSTV. (f) Ground truth.

$\eta_2 = 1.6 \cdot 10^{-4}$, $\eta_3 = 2.7 \cdot 10^{-5}$, $\mu = 2 \cdot 10^{-3}$. In addition, the parameters related to the testing image "Lemon slices" are as follows $\lambda = 0.7$, $\eta_1 = 3 \cdot 10^{-5}$, $\eta_2 = 1.8 \cdot 10^{-4}$, $\eta_3 = 2.3 \cdot 10^{-5}$, $\mu = 3 \cdot 10^{-3}$. At last, for the testing image "Flowers," we select the parameters as $\lambda = 0.7$, $\eta_1 = 1.6 \cdot 10^{-4}$, $\eta_2 = 3.3 \cdot 10^{-4}$, $\eta_3 = 3.2 \cdot 10^{-5}$, $\mu = 4 \cdot 10^{-3}$.

B. Weighting Methods Discussion

Here, we discuss the selection of the weighting vectors $\{\boldsymbol{\omega}^{(j)}\}$ in (10). To test the effectiveness of this approach, we utilize two images, "Superballs" and "Flowers" from the CAVE dataset. We denote the two different weighting schemes (32) and (33) as WNN1 and WNN2, respectively. The results of these tests can be found in Table I. The numerical results clearly indicate that WNN1 (32) is significantly less effective than WNN2 (33). These results demonstrate that the WNN2 weighting method is more suitable for HSI fusion and that the WNN1 method is not suitable for this task. These findings highlight the accuracy and practicality of the improved weighting method of WNN2.

C. Experimental Results

Table III presents the numerical results of five different fusion methods applied to four images. The effectiveness of a method is measured by the PSNR and UIQI indices, which should be high, and the ERGAS and SAM indices, which should be low. To facilitate comparison, the best value for each index is indicated in bold. Based on the table, it is evident that WNNSTV outperforms the other methods. Additionally, we compared the fusion results of the different methods

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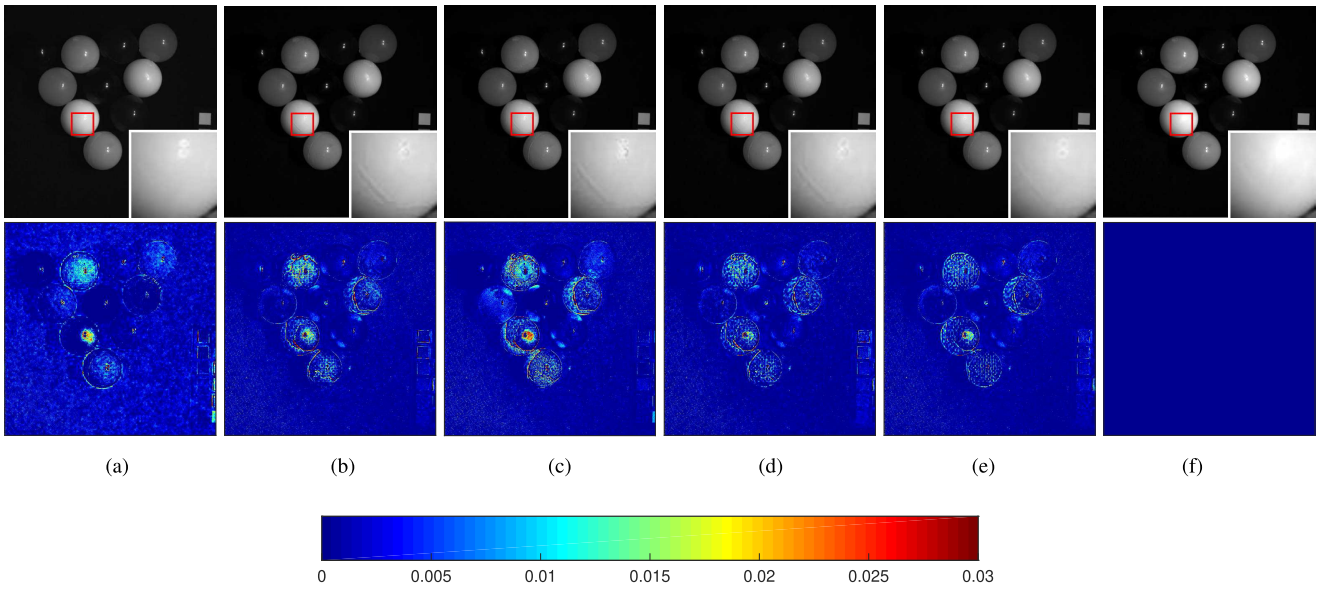


Fig. 3. Fused and error images of "Superballs" at the 21st spectral band. (a) HySure. (b) LRSR. (c) NSSJS. (d) SLRTV1. (e) WNNSTV. (f) Ground truth.

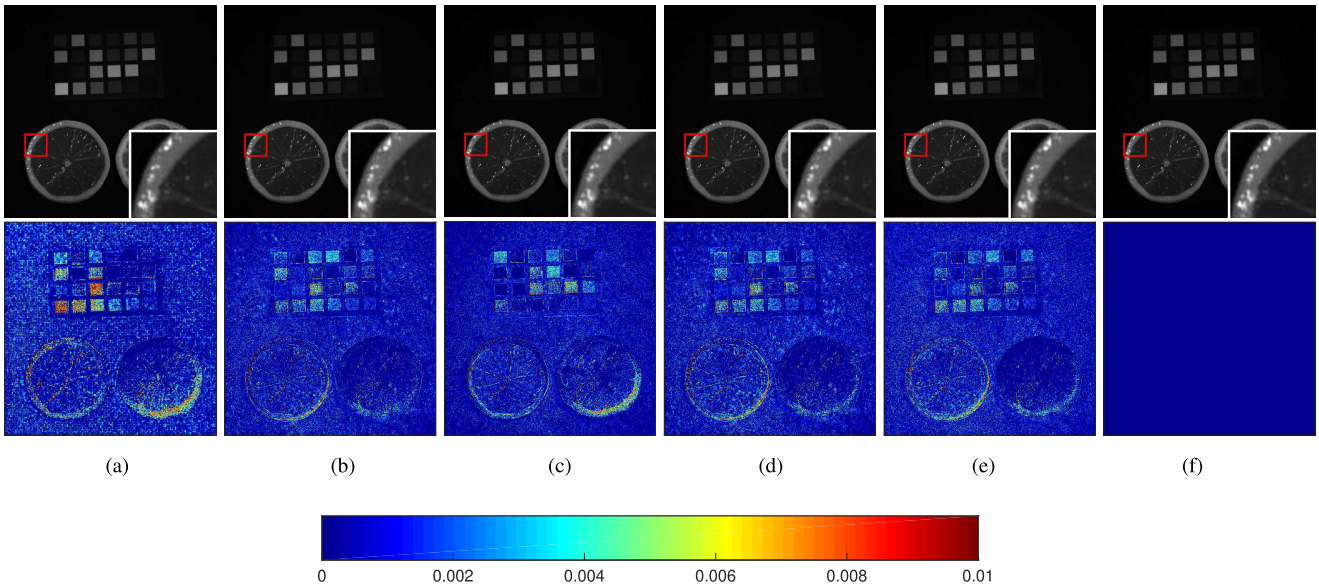


Fig. 4. Fused and error images of "Lemon slices" at the 25th spectral band. (a) HySure. (b) LRSR. (c) NSSJS. (d) SLRTV1. (e) WNNSTV. (f) Ground truth.

across various spectral bands, as shown in Fig. 1. The PSNR and UIQI curves of WNNSTV are higher than those of the other methods across almost all spectral bands, indicating its superiority. Specifically, the "Peppers" and "Flowers" images exhibit more pronounced improvements. Therefore, the experimental results demonstrate that WNNSTV yields superior fusion performance compared to the other methods.

In order to compare the effectiveness of various methods in preserving spatial structures, we present both fused and error images for each test image obtained by the different methods. The error image accurately reflects the disparity between the fused image and the ground truth. The results on the 20th spectral band of "Peppers" are shown in Fig. 2, showcasing both the fused images and the corresponding error images. The reconstructed images demonstrate that the WNNSTV method produces a smoother result compared to other methods, and gets a smaller disparity. This conclusion is further supported by the residual images, particularly for the pepper in the lower right-hand corner. Fig. 3 illustrates both the reconstructed images and the corresponding error images of "Superballs" captured at the 21st spectral band. By examining the enlarged subregions in Fig. 3, we observe that HySure overestimates brightness, while the other four methods produce spurious edges on the ball's surface. However, WNNSTV exhibits a superior fusion effect compared to other methods overall. Additionally, by examining the error images in Fig. 3, we observe that WNNSTV retains more spatial structural information on the surface of the balls. In terms of visual effect, the difference in fusion output between various methods is not easy to discern in the enlarged subregions presented in Fig. 4. However, upon examining the error images in Fig. 4, we notice that WNNSTV retains more

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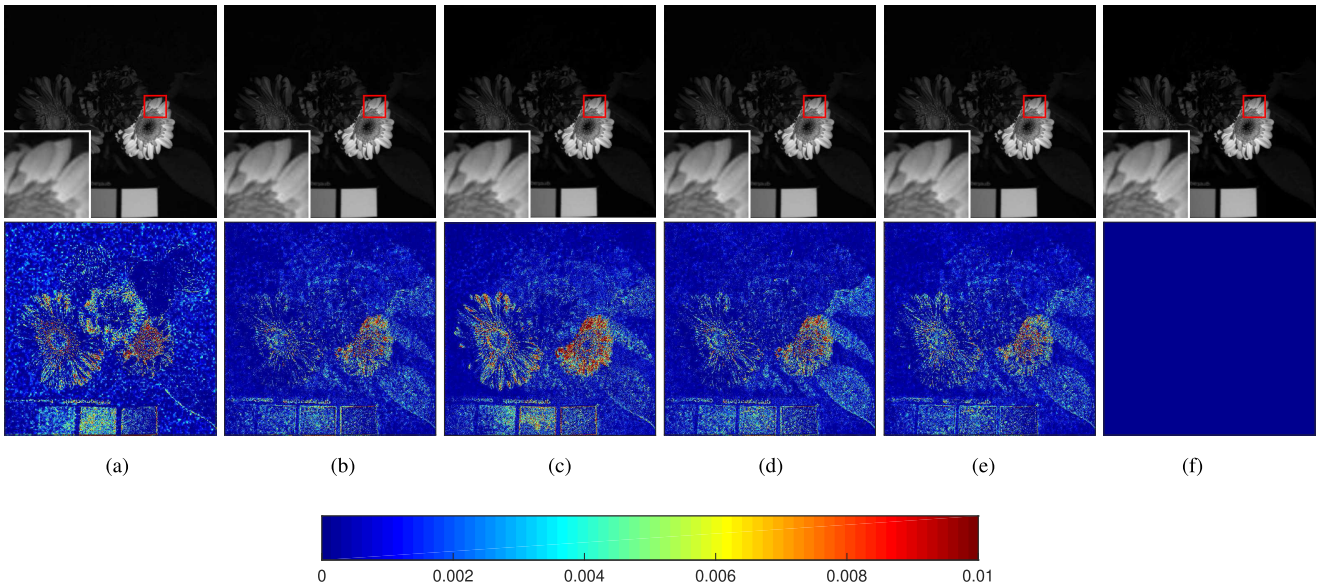


Fig. 5. Fused and error images of "Flowers" at the 18th spectral band. (a) HySure. (b) LRSR. (c) NSSJS. (d) SLRTV1. (e) WNNSTV. (f) Ground truth.

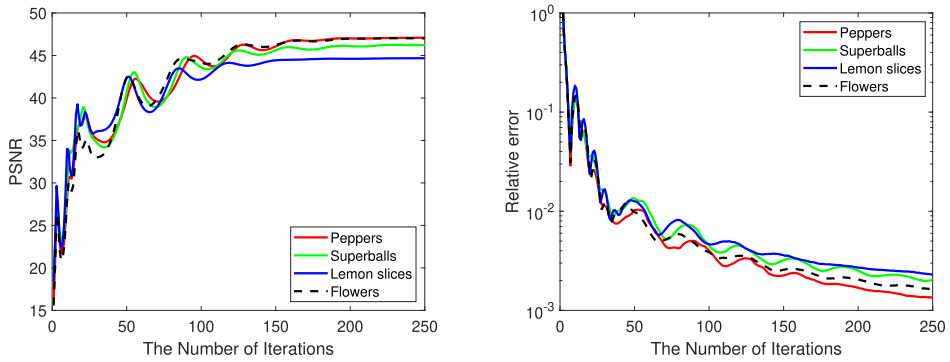


Fig. 6. Plots of PSNR and relative error versus the number of iterations for our WNNSTV. (a) PSNR. (b) Relative error.

details on the edges of lemon slices. The enlarged subregions shown in Fig. 5 demonstrate that the WNNSTV fusion output is visually clearer than those of other methods. Furthermore, by examining the error images in Fig. 5, we can observe that WNNSTV retains more structural details of flowers.

Finally, we experimentally demonstrate the convergence of Algorithm 1. Fig. 6 presents the plots of PSNR and relative error against the number of iterations for our WNNSTV. With an increase in the number of iterations, the PSNR values exhibit rapid growth, eventually stabilizing. Accordingly, their relative errors decrease rapidly. The observed trends provide substantial evidence of the proposed algorithm's convergence. It is noteworthy that the relative errors display initial oscillations during the early iterations, attributable to the nonconvex nature of WNN.

V. Conclusion

We introduced a novel hyperspectral sparse fusion model that incorporates WNN and TV regularization. The WNN assigns varying weights to singular values adaptively, thereby, better preserving major data components contained in larger singular values. An ADMM was utilized to solve the proposed model, where all subproblems had explicit solutions. Experimental results demonstrated the validity and superior performance of our approach. The future research directions involve improving the convergence rate of the proposed algorithm and extending the application of the WNN to other image processing tasks, including hyperspectral sparse unmixing and Poisson image restoration. In addition, we will study the properties of the proposed WNN model.

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